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Bioinformatics Approaches to Investigate Antimicrobial Resistance in *Staphylococcus aureus*

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ABSTRACT

Background:

Antimicrobial resistance (AMR) has become a major global health challenge, reducing the effectiveness of antibiotics and increasing the burden of bacterial infections. *Staphylococcus aureus*, particularly methicillin-resistant *S. aureus* (MRSA), is a clinically important pathogen that harbours multiple antimicrobial resistance determinants. This study employed an integrated bioinformatics workflow to investigate the genomic, transcriptomic, and functional characteristics of antimicrobial resistance in *S. aureus*.

Methodology:

Publicly available whole-genome sequencing and RNA sequencing datasets were obtained from the Sequence Read Archive (SRA), BV-BRC, and Gene Expression Omnibus (GEO). Genomic analyses included genome assembly, annotation, antimicrobial resistance gene identification, comparative genomics, phylogenetic reconstruction, pan-genome analysis, genomic context analysis, and protein functional characterization. Transcriptomic analyses involved quality assessment, read alignment, gene expression quantification, differential expression analysis, and functional annotation of differentially expressed genes. The results from these analyses were integrated to investigate antimicrobial resistance from multiple biological perspectives.

Results:

Comparative genomic analysis identified conserved core antimicrobial resistance genes, primarily associated with antibiotic efflux, together with variably distributed resistance determinants linked to methicillin resistance and disinfectant tolerance. Genomic context analysis showed that *mecA* was associated with SCCmec elements rather than predicted genomic islands, supporting the role of mobile genetic elements in the dissemination of methicillin resistance. Transcriptomic analysis identified differentially expressed genes associated with bacterial adaptation, while protein functional annotation and protein interaction analyses provided additional insight into the biological functions and relationships of representative resistance-associated proteins.

Conclusion:

The integration of comparative genomics, transcriptomics, and protein functional analyses provided a comprehensive framework for investigating antimicrobial resistance in *Staphylococcus aureus*. The integrated bioinformatics workflow developed in this study provides a systematic approach for genomic surveillance and future bioinformatics-based investigations of antimicrobial resistance in clinically important bacterial pathogens.

INTRODUCTION

Antimicrobial Resistance and *Staphylococcus aureus*

Antimicrobial resistance (AMR) has become one of the most significant challenges to global public health, reducing the effectiveness of antibiotics and limiting the treatment options available for bacterial infections. The widespread use and misuse of antimicrobial agents have accelerated the emergence and dissemination of resistant bacterial strains, resulting in increased morbidity, mortality, and healthcare costs worldwide. Understanding the genetic basis and mechanisms underlying antimicrobial resistance has therefore become a major

focus of microbiological and biomedical research (Guo et al., 2020; Mlynarczyk-Bonikowska et al., 2022).

Among bacterial pathogens, *Staphylococcus aureus* is of particular clinical importance because it is responsible for a wide range of infections, including skin and soft tissue infections, bacteremia, endocarditis, pneumonia, meningitis, and device-associated infections. As a member of the ESKAPE group of pathogens, *S. aureus* is able to survive under antimicrobial pressure and acquire resistance to multiple classes of antibiotics, making it a major cause of both community-acquired and hospital-acquired infections (Guo et al., 2020; Kaushik et al., 2024).

The emergence of methicillin-resistant *S. aureus* (MRSA) significantly changed the clinical management of staphylococcal infections. Methicillin resistance is primarily mediated by the *mecA* gene, which encodes the altered penicillin-binding protein PBP2a, reducing the effectiveness of β -lactam antibiotics (Guo et al., 2020; Mlynarczyk-Bonikowska et al., 2022; Naorem et al., 2020). In addition to *mecA*, numerous resistance genes contribute to resistance against other antimicrobial classes, and the accumulation of multiple resistance determinants has led to the emergence of multidrug-resistant strains that pose serious therapeutic challenges.

Advances in next-generation sequencing and bioinformatics have transformed the study of antimicrobial resistance by enabling large-scale analysis of genomic and transcriptomic data. Integrating these complementary datasets provides a more comprehensive understanding of the genetic organization, functional characteristics, and molecular mechanisms underlying antimicrobial resistance in *S. aureus*. Such integrated approaches support genomic surveillance and improve our understanding of the evolution and dissemination of antimicrobial resistance (Babiker et al., 2019; Mason et al., 2018; Pennone et al., 2022).

Bioinformatics Approaches to AMR Investigation

Advances in next-generation sequencing and bioinformatics have significantly improved the investigation of antimicrobial resistance by enabling comprehensive analysis of bacterial genomes and transcriptomes. Whole-genome sequencing allows the identification of antimicrobial resistance genes, sequence variations, and other genetic features associated with resistance, while publicly available genomic databases facilitate comparative analysis across bacterial isolates from different geographical regions and clinical settings (Babiker et al., 2019; Mason et al., 2018; Pennone et al., 2022).

Comparative genomics has become an important approach for studying the evolution of antimicrobial resistance in *Staphylococcus aureus*. Methods such as pan-genome characterization and phylogenetic reconstruction help distinguish conserved and variable genes, investigate genetic diversity, and examine the distribution of resistance determinants and mobile genetic elements among bacterial populations (John et al., 2019; Pennone et al., 2022; Souza et al., 2024).

Genomic information alone, however, does not indicate whether resistance-associated genes are actively expressed under different biological conditions. Transcriptomic analysis using RNA sequencing (RNA-seq) complements genomic studies by identifying changes in gene expression associated with bacterial adaptation (Love et al., 2014). Functional annotation approaches, including Gene Ontology enrichment, protein domain analysis, and protein

interaction networks, further assist in interpreting the biological significance of resistance-associated genes and proteins (**Gene Ontology Consortium, 2023; Hasan et al., 2025; Jones et al., 2014**).

Although each approach provides valuable information, no single method fully explains the complexity of antimicrobial resistance. Integrating comparative genomics, transcriptomics, and protein functional analyses offered a more comprehensive understanding of resistance mechanisms and formed the basis of the integrated bioinformatics workflow employed in this study (**Babiker et al., 2019; Hasan et al., 2025; Mason et al., 2018**).

Research Gap, Aim and Objectives

Previous studies have investigated antimicrobial resistance in *Staphylococcus aureus* using genomic, transcriptomic, or protein-level analyses. While these approaches have provided valuable insights into the identification of resistance genes, bacterial evolution, and functional characteristics, many studies have focused on individual analytical methods rather than integrating multiple bioinformatics approaches. Consequently, a comprehensive understanding of antimicrobial resistance requires the combined interpretation of genomic variation, gene expression, protein function, and genomic context (**Babiker et al., 2019; Hasan et al., 2025; Pennone et al., 2022**).

This study addresses this gap by implementing an integrated bioinformatics workflow that combines comparative genomics, antimicrobial resistance gene identification, transcriptomic analysis, protein functional annotation, protein interaction analysis and genomic context analysis to investigate antimicrobial resistance in *Staphylococcus aureus*. By integrating these complementary analyses, the study aims to provide a broader understanding of the genetic and functional mechanisms associated with antimicrobial resistance.

The primary aim of this study was to investigate antimicrobial resistance in *Staphylococcus aureus* using an integrated bioinformatics approach.

The specific objectives were to:

- Collect publicly available whole-genome sequencing and RNA sequencing datasets of *Staphylococcus aureus*.
- Identify and characterize antimicrobial resistance genes and their associated resistance mechanisms.
- Analyse differential gene expression using RNA sequencing datasets.
- Perform comparative genomic, phylogenetic, and pan-genome analyses to examine the distribution and evolution of antimicrobial resistance determinants.
- Functionally characterize selected antimicrobial resistance-associated proteins using protein annotation and interaction analyses.
- Integrate genomic, transcriptomic, and protein-level evidence to obtain a comprehensive understanding of antimicrobial resistance in *Staphylococcus aureus*.

METHODOLOGY

Study Design and Data Collection

This study aimed to investigate antimicrobial resistance (AMR) in *Staphylococcus aureus*, a clinically important human pathogen, using publicly available genomic and transcriptomic datasets. *Staphylococcus aureus* was selected as the study organism because of its role in both community- and hospital-acquired infections and the availability of well-characterized genomic and transcriptomic datasets in public repositories (Guo et al., 2020; Mlynarczyk-Bonikowska et al., 2022).

An integrated bioinformatics workflow was developed (**Figure 1**) to investigate AMR through comparative genomics, transcriptome analysis, and protein functional characterization. The workflow included sequence quality assessment, genome assembly, genome annotation, antimicrobial resistance gene identification, comparative genomic analysis, pangenome analysis, phylogenetic reconstruction, RNA-seq differential expression analysis, genomic context analysis, protein functional annotation, protein interaction analysis, and functional enrichment analysis. Together, these analyses provided complementary insights into the distribution, expression, evolution, and functional characteristics of AMR-associated genes.

Whole-genome sequencing data were obtained from two publicly available repositories: the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA; Leinonen et al., 2011) and the Bacterial and Viral Bioinformatics Resource Center (BV-BRC; formerly PATRIC; Olson et al., 2023; Wattam et al., 2017). Three raw sequencing datasets retrieved from the SRA were assembled and used as the primary study genomes for downstream genomic analyses. To support comparative genomic and pangenome analyses, thirteen complete or high-quality *S. aureus* reference genomes were downloaded from the PATRIC database. Following quality assessment and genome assembly, the three assembled genomes, together with the thirteen reference genomes, were included in the comparative genomic analyses.

Transcriptomic data were retrieved from the Gene Expression Omnibus (GEO; Edgar et al., 2002) database. A total of twenty RNA sequencing samples from three independent studies were selected to investigate gene expression under experimental conditions relevant to antimicrobial resistance. The datasets included both single-end and paired-end sequencing data and were processed using a standardized RNA-seq analysis pipeline to ensure consistency across all samples.

All computational analyses were performed in a Linux-based environment using open-source bioinformatics software. By integrating genomic, transcriptomic, comparative, and functional analyses, this study provides a comprehensive assessment of antimicrobial resistance in *Staphylococcus aureus* and highlights the value of combining multiple bioinformatics approaches to investigate resistance mechanisms.

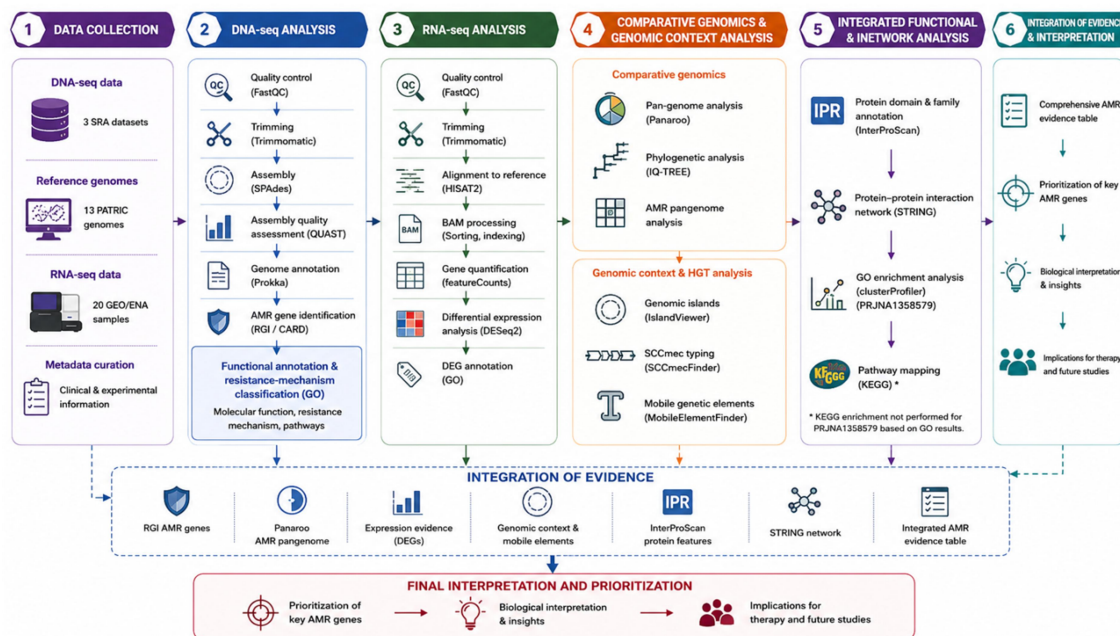


Figure 1: Overview of the bioinformatics workflow used in this study. The workflow includes genome assembly and quality assessment, antimicrobial resistance (AMR) gene identification, transcriptome analysis, comparative genomics, genomic context analysis, protein functional annotation, protein–protein interaction analysis, gene ontology enrichment, and integration of multiple lines of evidence to characterize AMR-associated genes in *Staphylococcus aureus*.

Whole Genome Sequence Analysis

Whole-genome sequencing data were processed to generate high-quality genome assemblies for subsequent antimicrobial resistance and comparative genomic analyses. Three raw whole-genome sequencing datasets retrieved from the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) were included in this phase of the study.

The quality of the raw sequencing reads was initially assessed using FastQC (**Andrews, 2010**) to evaluate parameters such as per-base sequence quality, GC content, sequence length distribution, adapter contamination, and sequence duplication levels. Adapter sequences and low-quality bases were removed using Trimmomatic (**Bolger et al., 2014**) to improve the overall quality of the sequencing reads. Following trimming, FastQC was performed again to verify the effectiveness of the quality filtering, and MultiQC (**Ewels et al., 2016**) reports were generated to summarise the quality metrics across all samples.

The trimmed paired-end reads were assembled into draft genomes using SPAdes (**Bankevich et al., 2012**), a de novo genome assembler designed for short-read sequencing data. The quality of the assembled genomes was assessed using QUAST (**Gurevich et al., 2013**) by examining assembly statistics including the number of contigs, total assembly length, N50 value, GC content, and other assembly quality metrics. This evaluation ensured that the assembled genomes were suitable for subsequent analyses.

Genome annotation was performed using Prokka (Seemann, 2014) to identify coding sequences (CDSs), transfer RNA (tRNA) genes, ribosomal RNA (rRNA) genes, and other genomic features within each assembled genome. The annotated genomes and predicted protein sequences generated during this step served as the basis for subsequent antimicrobial resistance gene identification and comparative genomic analyses. Representative AMR protein sequences identified in later phases were further used for protein functional annotation.

AMR Gene Identification and Functional Analysis

Following genome annotation, antimicrobial resistance (AMR) genes were identified from all analysed genomes using the Resistance Gene Identifier (RGI). RGI was used in conjunction with the Comprehensive Antibiotic Resistance Database (CARD ; Alcock et al., 2023) to identify known resistance genes and resistance-associated mutations based on curated reference sequences and resistance models.

The RGI output from each genome was compiled to generate a comprehensive catalogue of AMR genes present across the analysed genomes. The identified genes were examined for their occurrence in different genomes and classified according to the resistance mechanisms assigned by CARD. These included mechanisms such as antibiotic efflux, antibiotic target alteration, antibiotic target replacement, antibiotic inactivation, antibiotic target protection, and reduced permeability to antibiotics. Drug class annotations provided by CARD were also recorded to identify the classes of antimicrobial agents associated with each resistance gene.

To further characterise the identified AMR genes, Gene Ontology (GO; **Gene Ontology Consortium, 2023**) and Kyoto Encyclopedia of Genes and Genomes (KEGG; **Kanehisa et al., 2023**) annotations were compiled for selected priority resistance genes. GO annotations were used to describe their molecular functions and biological processes, while KEGG pathway information was used to summarise the biological pathways and resistance-related functions associated with these genes.

The annotated AMR gene catalogue generated during this phase was subsequently used for comparative genomic analyses, selection of representative proteins for functional annotation and integration of genomic and transcriptomic evidence in the later stages of the study.

RNA-seq Analysis

RNA sequencing (RNA-seq) analysis was performed to investigate differential gene expression associated with antimicrobial resistance in *Staphylococcus aureus*. A total of twenty RNA sequencing samples representing three independent studies were retrieved from the Gene Expression Omnibus (GEO) database. The datasets included both single-end and paired-end sequencing reads and were processed using a standardized analysis pipeline.

The quality of the raw RNA-seq reads was initially assessed using FastQC to evaluate sequencing quality, GC content, sequence length distribution, adapter contamination, and other quality metrics. Adapter trimming and quality filtering were carried out using Trimmomatic for datasets requiring preprocessing. MultiQC was then used to compile the FastQC reports and provide an overall assessment of sequencing quality across all samples.

The reference genome of *Staphylococcus aureus* N315 (RefSeq accession GCF_000009645.1; O’Leary et al., 2016) was indexed using HISAT2 (Kim et al., 2019) prior to read alignment. Quality-filtered reads were aligned to the *Staphylococcus aureus* N315 reference genome using HISAT2, and the resulting Sequence Alignment/Map (SAM) files were converted to Binary Alignment/Map (BAM) format and indexed using SAMtools (Li et al., 2009). Gene-level read counts were generated from the aligned BAM files using featureCounts (Liao et al., 2014) based on the annotated reference genome.

Differential gene expression analysis was performed using the DESeq2 (Love et al., 2014) package in R (R Core Team, 2026). Gene expression levels between the experimental conditions within each dataset were compared to identify significantly differentially expressed genes. The resulting differentially expressed genes were subsequently annotated using the reference genome annotation to obtain gene identifiers, locus tags, and functional descriptions for downstream biological interpretation. Functional annotation of the differentially expressed genes was subsequently performed to facilitate biological interpretation and downstream enrichment analyses.

Comparative Genomics

Comparative genomic analysis was performed to investigate the conservation and distribution of antimicrobial resistance (AMR) genes among the analysed *Staphylococcus aureus* genomes. The annotated genomes generated during the previous phases, including the assembled genomes from the SRA datasets and the reference genomes obtained from the Bacterial and Viral Bioinformatics Resource Center (BV-BRC), were used for this analysis.

Pangenome analysis was carried out using Panaroo (Tonkin-Hill et al., 2020) to identify homologous gene clusters across all genomes and classify them into core, soft-core, shell, and cloud components of the *S. aureus* pangenome. Panaroo also generated a core genome multiple sequence alignment, which was used for subsequent phylogenetic analysis.

Phylogenetic reconstruction was performed using IQ-TREE (Minh et al., 2020) based on the core genome alignment produced by Panaroo. The resulting phylogenetic tree was used to examine the evolutionary relationships among the analysed genomes and to provide a genomic framework for interpreting the distribution of AMR genes.

To investigate the conservation of resistance determinants, the AMR genes identified using CARD-RGI were compared with the Panaroo gene presence–absence matrix. This comparison was used to determine whether individual resistance genes were conserved across all genomes or restricted to a subset of genomes. Based on this comparison, an AMR pangenome table was constructed by integrating resistance gene information with pangenome classifications, providing a comprehensive summary of the distribution of AMR-associated genes across the analysed genomes.

Genomic Context Analysis

Genomic context analysis was performed to investigate the genomic environment and potential mobility of antimicrobial resistance (AMR) genes identified in the assembled *Staphylococcus aureus* genomes. Since complete genome assemblies were generated only for the whole-genome sequencing datasets obtained from the NCBI Sequence Read Archive (SRA), these analyses were carried out on the three assembled genomes.

Genomic islands were identified using IslandViewer (**Bertelli et al., 2017**), which integrates multiple prediction methods to detect regions acquired through horizontal gene transfer. The predicted genomic islands were examined to determine whether they contained antimicrobial resistance-associated genes or other genomic features associated with bacterial adaptation and antimicrobial resistance.

The chromosomal cassette carrying methicillin resistance was characterised using SCCmecFinder (**Kaya et al., 2018**). This analysis was performed to identify the SCCmec type present in each assembled genome and to characterise the genomic context of the *mecA* gene, which is responsible for methicillin resistance in *S. aureus*.

Mobile genetic elements were identified using MobileElementFinder (**Johansson et al., 2021**) to detect insertion sequences, transposons, and other mobile DNA elements that may contribute to the acquisition and dissemination of antimicrobial resistance genes. The combined results from IslandViewer, SCCmecFinder, and MobileElementFinder were used to assess the genomic context of the identified AMR genes and to evaluate evidence supporting horizontal gene transfer.

Protein Functional Annotation

Protein functional annotation was performed to investigate the structural and functional characteristics of selected antimicrobial resistance (AMR) proteins identified during the genomic analysis. Representative protein sequences were selected from the CARD-RGI results for seven priority AMR genes based on their biological significance and distribution among the analysed genomes.

The selected protein sequences were compiled into a FASTA file and analysed using InterProScan (**Jones et al., 2014**). Multiple integrated protein signature databases within InterProScan were used to identify conserved protein families, functional domains, and other sequence signatures associated with each AMR protein. Gene Ontology (GO) terms generated by InterProScan were also retrieved to provide additional functional information regarding the molecular functions and biological processes of the annotated proteins.

The InterProScan output was manually reviewed to select representative annotations for each AMR protein. Information including protein family, conserved domains, InterPro identifiers, GO terms and membrane-associated features was compiled into an InterPro Summary Table. This summary provided a standardized functional annotation for the representative AMR proteins and was subsequently used for the integration of genomic, transcriptomic, and protein-level evidence.

Data Integration

The final phase of the study involved the integration of genomic, transcriptomic, and protein-level information to obtain a comprehensive overview of antimicrobial resistance (AMR) in *Staphylococcus aureus*. Results generated from the previous analyses were systematically combined to facilitate biological interpretation of the identified resistance genes.

An Integrated AMR Evidence Table was constructed by combining information obtained from CARD-RGI, Panaroo, RNA-seq differential expression analysis, InterProScan, SCCmecFinder, IslandViewer, and MobileElementFinder. For each priority AMR gene, the

table summarised the resistance mechanism, pangenome classification, protein functional annotation, differential expression status, genomic context, and association with mobile genetic elements. This integrated dataset enabled comparison of evidence from multiple analytical approaches within a single framework.

Protein–protein interaction analysis was performed using the STRING (**Szklarczyk et al., 2023**) database to investigate the functional relationships among the selected priority AMR genes. The interaction network was analysed to identify functional associations among the selected AMR proteins based on evidence integrated within the STRING database.

To investigate the biological processes associated with transcriptional changes, Gene Ontology (GO) enrichment analysis was performed using the clusterProfiler (**Wu et al., 2021**) package in R. The analysis was carried out on the upregulated differentially expressed genes identified from the PRJNA1358579 RNA-seq dataset, which contained the highest number of differentially expressed genes among the analysed datasets. Statistically enriched GO terms were identified after multiple testing correction and were used to characterise the biological functions associated with the observed transcriptional response.

The integrated datasets generated during this phase were used to support the biological interpretation of AMR-associated genes by combining evidence from genome analysis, comparative genomics, transcriptomics, protein functional annotation, and genomic context analysis.

RESULTS

Genome Assembly Quality Assessment

The quality of the three *Staphylococcus aureus* genome assemblies was assessed using QUAST (**Figure 2**). The assembled genome sizes ranged from **2.78 Mb to 2.82 Mb**, with GC contents between **32.6% and 32.7%**, indicating similar genome characteristics across the three isolates.

Assembly continuity was evaluated using the N50 statistic and the number of contigs. Among the assemblies, **ERR13347068** showed the highest continuity, with an N50 of **220,588 bp** and the lowest number of contigs (**27**). **ERR12712640** had an N50 of **168,434 bp** with **34** contigs, whereas **ERR12712641** had an N50 of **154,095 bp** and **37** contigs.

The QUAST results showed that all three assemblies were of good quality and were suitable for subsequent analyses, including genome annotation, antimicrobial resistance gene identification, comparative genomics, and genomic context analysis.

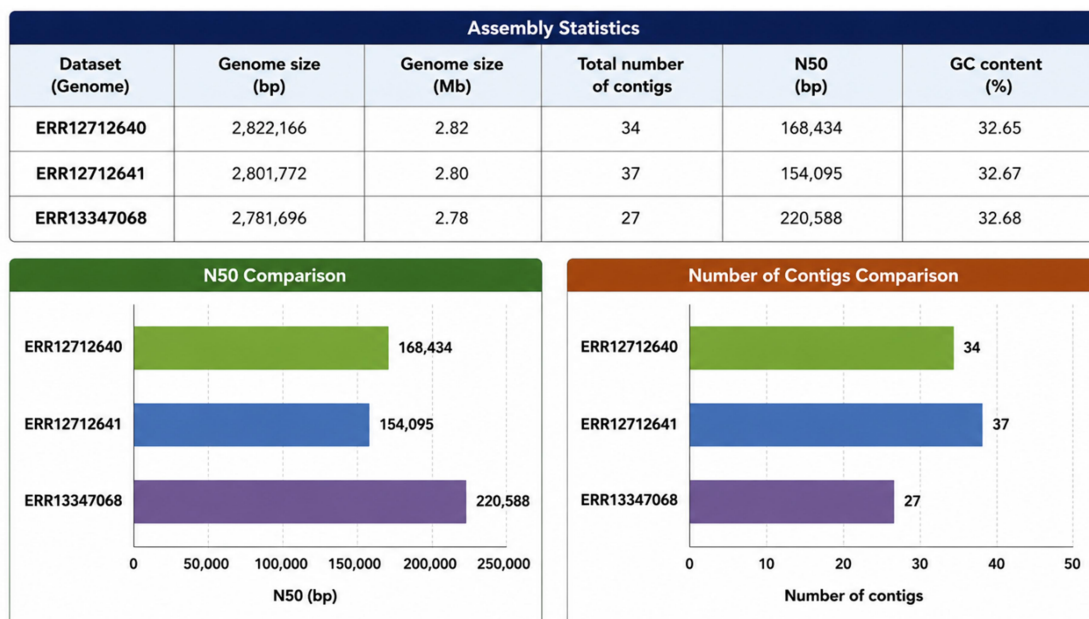


Figure 2: Quality assessment of the three *Staphylococcus aureus* genomes assembled from SRA sequencing data using QUAST. The comparison summarizes key assembly statistics, including genome size, GC content, number of contigs, N50 value, and predicted coding sequences, demonstrating the quality of the assembled genomes used for downstream analyses.

AMR Gene Distribution Across *Staphylococcus aureus* Genomes

The distribution of antimicrobial resistance (AMR) genes across the sixteen analysed *Staphylococcus aureus* genomes is shown in the presence–absence heatmap (**Figure 3**). Several AMR genes were detected in nearly all analysed genomes, whereas others were present only in selected isolates.

Genes such as *mepA*, *mepR*, *tet(38)*, *norC*, *sepA* (small multidrug resistance (SMR) antibiotic efflux pump) and the *vanT* gene cluster were detected in most of the analysed genomes. *blaZ* (BlaZ beta-lactamase) and *fosB* were detected in a majority of the genomes, while *mecA* (methicillin resistant PBP2), *gyrA*, *parC*, *ANT(4')*, *ANT(6)*, *ANT(9)*, *APH(3')*, *dfr*, and *SAT* were identified only in a minority of the genomes.

These results show that the sixteen *S. aureus* genomes shared a core set of AMR genes, while several resistance determinants exhibited isolate-specific presence–absence patterns.

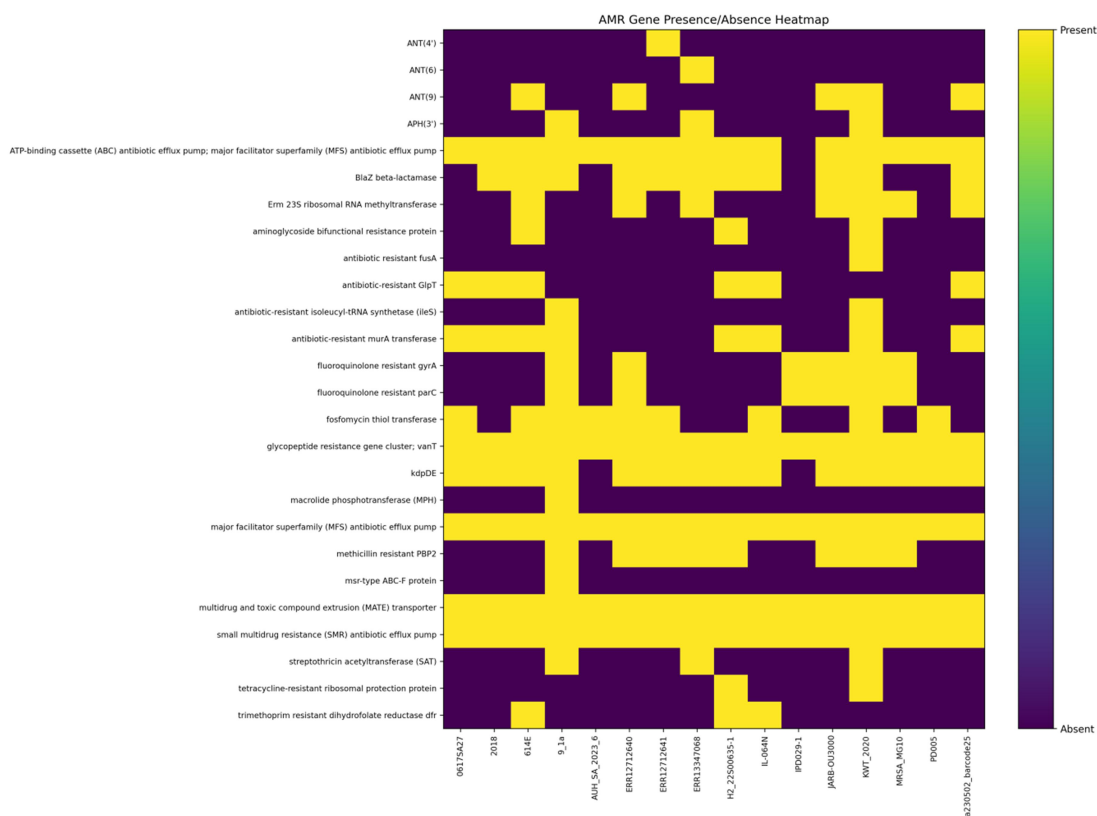


Figure 3: Distribution of representative antimicrobial resistance genes across the sixteen *Staphylococcus aureus* genomes analyzed in this study. The heatmap illustrates the presence or absence of selected AMR genes identified using the Comprehensive Antibiotic Resistance Database (CARD) and Resistance Gene Identifier (RGI).

Distribution of Resistance Mechanisms Among Representative AMR Genes

The resistance mechanisms associated with the seven representative antimicrobial resistance (AMR) genes selected for detailed functional analysis are shown in **Figure 4**. The selected genes were **mecA**, **norA**, **mepA**, **mepR**, **tet(38)**, **gyrA**, and **parC**.

Among these genes, **antibiotic efflux** was the most common resistance mechanism and was associated with **norA**, **mepA**, and **tet(38)**. **Target alteration** was observed for **gyrA** and **parC**, whereas **regulation of antibiotic efflux** and **antibiotic target replacement** were represented by **mepR** and **mecA**, respectively.

The seven representative AMR genes were classified into four distinct resistance mechanisms, with antibiotic efflux being the most frequently represented category.

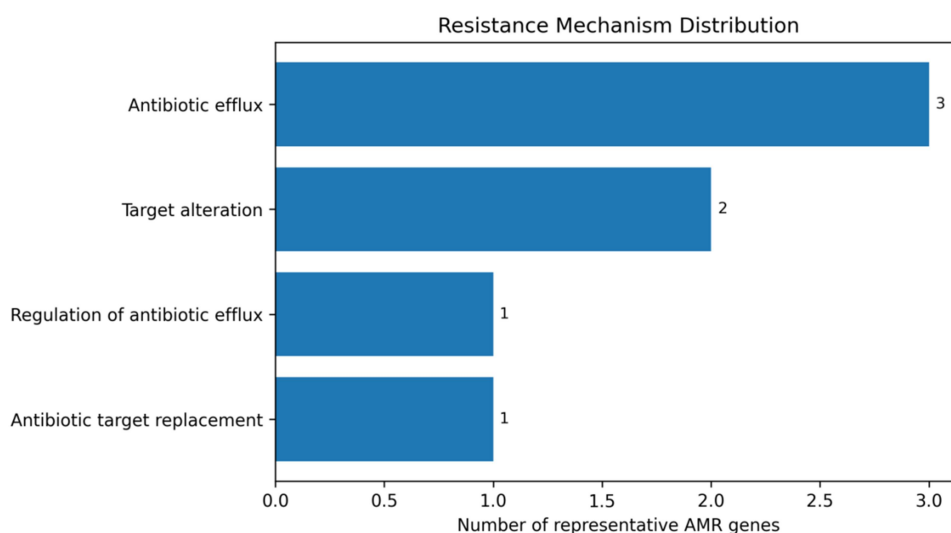


Figure 4: Distribution of resistance mechanisms among the seven representative antimicrobial resistance genes selected for downstream functional analyses. Antibiotic efflux was the most frequently represented mechanism, followed by target alteration, antibiotic target replacement, and regulation of antibiotic efflux.

Phylogenetic Analysis of *Staphylococcus aureus* Genomes

The evolutionary relationships among the sixteen *Staphylococcus aureus* genomes were inferred using a maximum-likelihood phylogenetic tree constructed from the Panaroo core genome alignment (**Figure 5**). The tree grouped the analysed genomes into several distinct clades.

The three assembled genomes obtained from the SRA datasets were distributed across two different clades. **ERR12712640** and **ERR12712641** clustered together with a bootstrap support value of **100**, indicating a close evolutionary relationship between these two isolates. In contrast, **ERR13347068** clustered with the reference genome **JARB-OU3000**, also with a bootstrap support value of **100**.

Several additional closely related genome pairs were observed, including **2018** and **H2_22S00635-1**, **614E** and **sa230502_barcode25**, and **MRSA_MG10** and

AUH_SA_2023_6, each supported by high bootstrap values. Other genomes, including **9_1a**, **IPD029-1**, **KWT_2020**, and **PD005**, formed separate branches within the phylogeny.

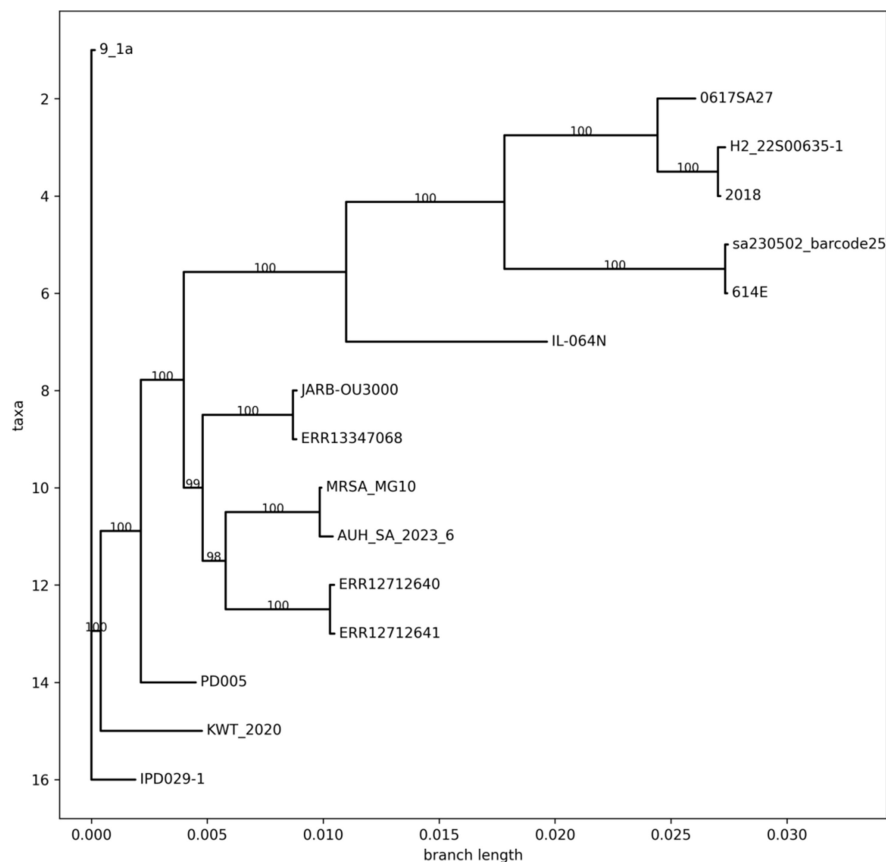


Figure 5: Maximum-likelihood phylogenetic tree constructed from the core genome alignment of sixteen *Staphylococcus aureus* genomes using IQ-TREE. The tree illustrates the evolutionary relationships among the assembled SRA genomes and the BV-BRC reference genomes included in the comparative genomic analysis.

Distribution of Representative AMR Genes in the *Staphylococcus aureus* Pangenome

The distribution of the seven representative antimicrobial resistance (AMR) genes across the sixteen analysed *Staphylococcus aureus* genomes is shown in the AMR pangenome heatmap (**Figure 6**). The heatmap illustrates the presence and absence of each representative gene based on the Panaroo gene presence–absence analysis.

Among the seven genes, **norA**, **mepA**, **mepR**, **parC**, and **tet(38)** were present in most of the analysed genomes. In contrast, **mecA** and **gyrA** were detected only in a subset of the genomes.

Differences in the representative AMR gene profiles were observed among individual genomes, with some genomes containing all or most of the selected genes, whereas others lacked one or more genes, particularly **mecA** and **gyrA**.

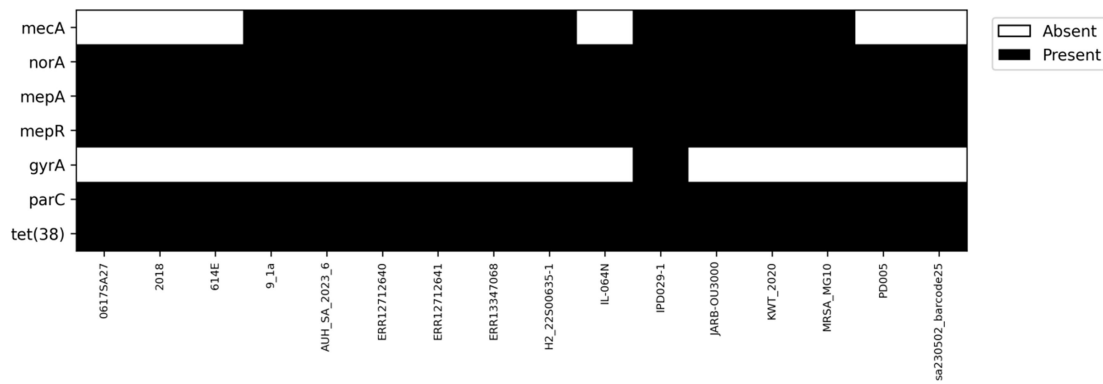


Figure 6: Distribution of the seven representative antimicrobial resistance genes within the *Staphylococcus aureus* pangenome. The heatmap summarizes the presence or absence of each representative gene across the sixteen analyzed genomes based on Panaroo pangenome clustering and integrated evidence.

Genomic Context Analysis of the Assembled Genomes

The genomic context of the three assembled *Staphylococcus aureus* genomes was analysed using IslandViewer, SCCmecFinder, and MobileElementFinder (**Figure 7**).

IslandViewer predicted **nine genomic islands** in both **ERR12712640** and **ERR12712641**, whereas **ERR13347068** contained **six** predicted genomic islands. No antimicrobial resistance or virulence genes were identified within the predicted genomic islands in any of the three genomes.

SCCmecFinder identified different SCCmec types in the assembled genomes. **ERR12712640** carried **SCCmec type II (2A)**, **ERR12712641** carried **SCCmec type IVc (2B)**, and **ERR13347068** carried **SCCmec type IVa (2B)**.

MobileElementFinder identified multiple mobile genetic elements in each genome. **ERR12712640** contained **ISSau4**, **ISSau3**, **ISSau6**, and **Tn554**. **ERR12712641** contained **three copies of ISSau6**, **Tn552**, and the composite transposon **cn_6077_ISSau6**, whereas **ERR13347068** contained **ISSep3** and **Tn5405**.

IslandViewer, SCCmecFinder and MobileElementFinder summary

ERR12712640	ERR12712641	ERR13347068
Predicted genomic islands: 9	Predicted genomic islands: 9	Predicted genomic islands: 6
AMR/VF genes in islands: None detected	AMR/VF genes in islands: None detected	AMR/VF genes in islands: None detected
SCCmec: Type II (2A)	SCCmec: Type IVc (2B)	SCCmec: Type IVa (2B)
Mobile genetic elements	Mobile genetic elements	Mobile genetic elements
ISSau4 ISSau3 ISSau6 Tn554	ISSau6 (3 copies) Tn552 Composite transposon (cn_6077_ISSau6)	ISSep3 Tn5405

Figure 7: Integrated genomic context analysis of the three assembled *Staphylococcus aureus* genomes using IslandViewer, SCCmecFinder, and MobileElementFinder. The figure summarizes the number of predicted genomic islands, SCCmec types, and detected mobile genetic elements for each genome. No antimicrobial resistance or virulence genes were identified within the predicted genomic islands.

Protein–Protein Interaction Network of Representative AMR Genes

Protein–protein interactions among the representative antimicrobial resistance (AMR) genes were examined using the STRING database (**Figure 8**). Five of the seven representative proteins were recognised in the selected *Staphylococcus aureus* reference proteome, while two proteins could not be mapped. The **norA** protein was mapped to the corresponding reference protein annotation (**norG**) used by STRING.

A single interaction was identified between **gyrA** and **parC**, with a combined STRING interaction score of **0.753**. No direct interactions were identified for **mecA**, **mepA**, or **norG**, which appeared as isolated nodes in the network.

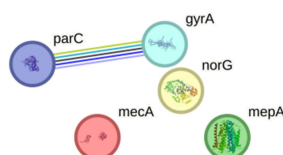


Figure 8: Protein–protein interaction network of representative antimicrobial resistance-associated proteins generated using STRING. Nodes represent proteins and edges indicate predicted functional associations. A strong interaction was observed between **gyrA** and **parC**, whereas **mecA**, **mepA**, and **norG** were not connected under the selected confidence threshold.

Gene Ontology Enrichment Analysis

Gene Ontology (GO) enrichment analysis was performed on the upregulated genes identified from the PRJNA1358579 RNA-seq dataset (**Figure 9**). Three GO terms were significantly enriched after multiple testing correction.

The terms **structural constituent of ribosome** (GO:0003735) and **translation** (GO:0006412) showed the highest enrichment, each with a **GeneRatio of 6/13**, fold enrichment values of **12.02** and **11.22**, respectively, and an adjusted p -value of **3.11×10^{-5}** . The GO term **ribosome** (GO:0005840) was also significantly enriched, with a **GeneRatio of 3/13**, a fold enrichment of **10.20**, and an adjusted p -value of **0.0104**.

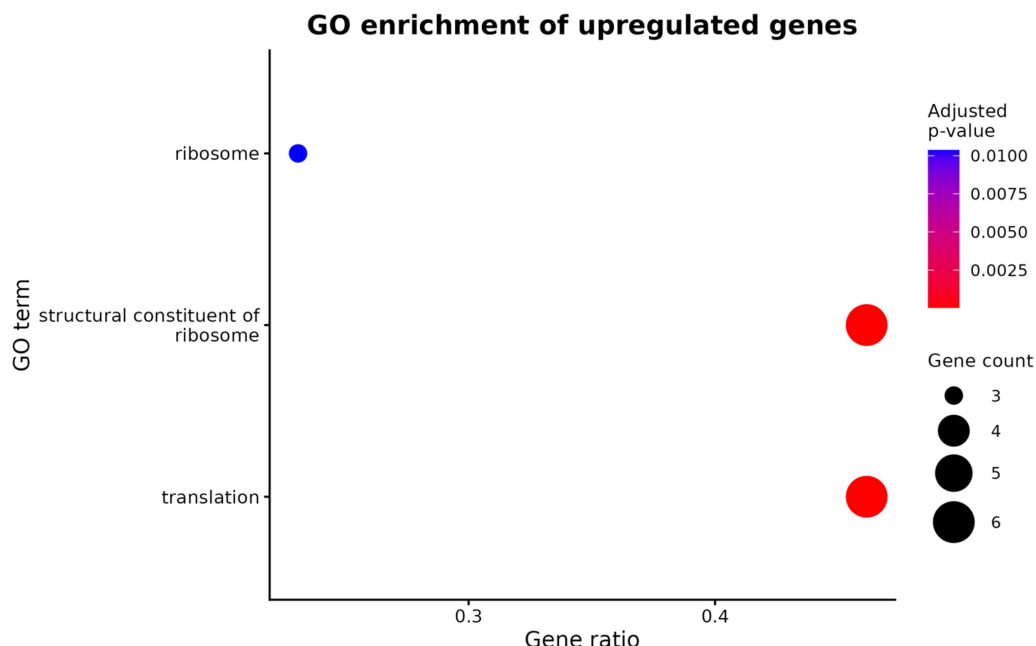


Figure 9: Gene Ontology enrichment analysis of significantly upregulated genes identified in the PRJNA1358579 RNA-seq dataset. The dot plot displays enriched biological functions according to gene ratio, gene count, and adjusted p -value, highlighting enrichment of ribosome-associated and translation-related processes.

Integrated AMR Evidence Analysis

The integrated AMR evidence matrix combines resistance annotation, pangenome distribution, RNA-seq differential expression, protein functional annotation, and genomic context information for the seven representative AMR genes (**Figure 10**).

Panaroo classified **norA**, **mepA**, **mepR**, **gyrA**, and **parC** as core genes present in all sixteen genomes, whereas **mecA** was classified as a shell gene and was present in **10/16** genomes. **tet(38)** was detected by RGI in all sixteen genomes but was not listed in the Panaroo output.

No significant differential expression was recorded for **norA**, **mepA**, **mepR**, **tet(38)**, **gyrA**, or **parC** in the RNA-seq comparisons included in the matrix. **mecA** was also not identified as differentially expressed, although **psm-mec** was upregulated in the GSE40864 dataset.

InterProScan annotation identified **mecA** as a PBP2a transpeptidase, **norA** and **tet(38)** as MFS-associated efflux proteins, **mepA** as a MATE efflux transporter, **mepR** as a MarR-family transcriptional regulator, and **gyrA** and **parC** as DNA topoisomerase proteins.

SCCmec association was recorded for **mecA** in all three SRA-derived genomes subjected to SCCmec typing. None of the seven representative genes was detected within the genomic islands examined, and no direct insertion sequence or transposon association was recorded for the remaining representative genes in the integrated matrix.

Gene	Resistance (RGI)	Pangenome (Panaroo)	RNA-seq DEGs	Functional Annotation (InterProScan)		Genomic Context Evidence			Overall Interpretation
	Mechanism / drug class / RGI detection	Status	Expression	Family	Key domain	SCCmec Association	Genomic Island	Mobile Element	Role in AMR
 mecA	Target replacement (PBP2a); β -lactam; 8/16	Shell (10/16)	Not DEG; psm-mec $\uparrow$ GSE40864	PBP2a transpeptidase family	PBP transpeptidase domain	✓ Present, all 31 typed genomes	Not detected	SCCmec-linked; no IS/Tn	Major methicillin-resistance determinant; SCCmec cassette-mediated mobility
 norA	Efflux (MFS); fluoroquinolone; 15/16	Core (16/16)	No significant DE	MFS efflux transporter	MFS domain (PF07690)	— Not applicable (core gene)	— Not detected	None detected	Core housekeeping efflux pump
 mepA	Efflux (MATE); tetracycline; 16/16	Core (16/16)	No significant DE	MATE efflux transporter	MatE domain (PF01554)	— Not applicable (core gene)	— Not detected	None detected	Core baseline efflux pump
 mepR	Efflux regulator; represses <i>mepA</i> ; 16/16	Core (16/16)	No significant DE	MarR transcriptional regulator	MarR DNA-binding domain	— Not applicable (core gene)	— Not detected	None detected	Core repressor of <i>mepA</i>
 tet(38)	Efflux (MFS); tetracycline; 16/16	Not listed (Panaroo)	No significant DE	MFS tetracycline efflux (Tet38)	MFS domain (PF07690)	— Not applicable (core gene)	— Not detected	None detected	RGI-only; likely Panaroo gap
 gyrA	Target alteration; SNP S84L; 6/16	Core (16/16)	No significant DE	DNA gyrase subunit A	Gyrase/topo IV domain (PF00521)	— Not applicable (core gene)	— Not detected	None (point mutation)	Core gene; QRDR mutation
 parC	Target alteration; SNPs S80F/E84; 6/16	Core (16/16)	No significant DE	Topoisomerase IV subunit A	Gyrase/topo IV domain (PF00521)	— Not applicable (core gene)	— Not detected	None (point mutation)	Core gene; co-occurs with <i>gyrA</i>

Figure 10: Integrated evidence matrix summarizing multiple lines of evidence for the seven representative antimicrobial resistance genes. The matrix combines resistance mechanisms identified by RGI, protein functional annotations from InterProScan, pangenome classification from Panaroo, RNA-seq differential expression, genomic context information from SCCmecFinder, IslandViewer, and MobileElementFinder, and the overall interpretation for each gene, providing a comprehensive overview of their roles in antimicrobial resistance in *Staphylococcus aureus*.

DISCUSSION

Overview of the Major Findings

This study applied an integrated bioinformatics approach to investigate antimicrobial resistance (AMR) in *Staphylococcus aureus* by combining comparative genomics, transcriptome analysis, protein functional annotation, and mobile genetic element analysis. Rather than relying on a single analytical strategy, the study examined AMR from multiple biological perspectives to better understand the distribution, functional characteristics, and potential regulation of resistance determinants. The integration of genomic and transcriptomic data provided a more comprehensive understanding of AMR by revealing not only which resistance genes were present in the genomes but also how bacterial gene expression and protein function may contribute to antimicrobial resistance.

The comparative genomic analyses showed that several AMR-associated genes were highly conserved among the studied genomes, indicating that they constitute stable components of the *S. aureus* genome, whereas other resistance determinants exhibited a more variable distribution across isolates. The identification of SCCmec elements carrying methicillin resistance genes supports the established role of these mobile genetic elements in the dissemination of β -lactam resistance (Kaya et al., 2018; Mlynarczyk-Bonikowska et al., 2022; Naorem et al., 2020). Although the analyzed resistance genes were not located within the genomic islands predicted by IslandViewer, the detection of insertion sequences and

transposable elements suggests a potential role for horizontal gene transfer in the evolution and dissemination of antimicrobial resistance (**John et al., 2019; Souza et al., 2024**).

The transcriptomic analyses complemented these genomic findings by providing insight into the functional response of *S. aureus* under different experimental conditions. Functional enrichment indicated that differentially expressed genes were primarily associated with protein synthesis and translational processes, suggesting adaptive changes in cellular activity under the investigated conditions. Protein domain analysis further confirmed the conserved structural features of representative AMR proteins, while protein interaction analysis identified a functional association between **gyrA** and **parC**, whereas the remaining representative proteins did not form direct interactions in the selected reference proteome. Overall, integrating comparative genomics, transcriptomics, and protein functional analyses provided complementary evidence for understanding the genetic and functional basis of antimicrobial resistance in *S. aureus*.

Comparative Genomics Reveals Conserved and Variable Antimicrobial Resistance Determinants

Comparative genomic analysis demonstrated that antimicrobial resistance in *Staphylococcus aureus* is supported by a combination of conserved intrinsic resistance determinants and variably distributed acquired resistance genes. The identification of several AMR-associated genes as components of the core genome suggests that these genes represent stable features of the species and may contribute to the basal resistance phenotype observed in many *S. aureus* isolates. Many of these genes encode multidrug efflux systems or their associated regulators, which are known to reduce intracellular antibiotic concentrations and contribute to reduced susceptibility to multiple classes of antimicrobial agents (**Guo et al., 2020; Mlynarczyk-Bonikowska et al., 2022**). The widespread conservation of these genes across the analysed genomes suggests that efflux-mediated resistance forms an important component of the intrinsic resistome of *S. aureus*.

In contrast, the variable distribution of genes such as *mecA* and other SCCmec-associated determinants reflects the dynamic nature of acquired antimicrobial resistance. The presence of these genes in only a subset of genomes is consistent with previous studies showing that methicillin resistance is primarily mediated through the acquisition of SCCmec rather than being an inherent characteristic of all *S. aureus* strains (**Kaya et al., 2018; Mlynarczyk-Bonikowska et al., 2022; Naorem et al., 2020**). This distinction between conserved and acquired resistance mechanisms highlights the importance of comparative genomics for differentiating species-level resistance traits from lineage-specific adaptations.

The genomic context analyses further supported this interpretation. SCCmecFinder identified SCCmec elements in the assembled genomes, supporting the association between *mecA* and this mobile genetic element. Although IslandViewer did not predict genomic islands containing the analyzed resistance genes, MobileElementFinder detected multiple insertion sequences and transposable elements within the assembled genomes. These findings suggest a potential role for mobile genetic elements in facilitating genome evolution and the dissemination of antimicrobial resistance genes, even when the resistance determinants themselves are not located within predicted genomic islands. Together, these findings indicate that both conserved genomic features and mobile genetic elements contribute to the diversity of antimicrobial resistance observed among the analysed *S. aureus* genomes.

Functional Insights from Transcriptomic and Protein-Level Analyses

While comparative genomics identified the repertoire of antimicrobial resistance genes present in *Staphylococcus aureus*, transcriptomic and protein-level analyses provided additional insight into their potential functional significance. Gene expression analysis demonstrated that the three RNA-seq datasets exhibited distinct transcriptional responses under their respective experimental conditions, with PRJNA1358579 producing the largest set of differentially expressed genes and therefore serving as the basis for downstream functional enrichment analysis. This observation indicates that transcriptional responses varied across the different experimental conditions, emphasizing the value of integrating transcriptomic data with genomic analyses.

GO enrichment analysis of the differentially expressed genes from PRJNA1358579 identified significant enrichment of translation- and ribosome-related biological processes. Rather than indicating the activation of specific antimicrobial resistance pathways, these results suggest that the bacterial response under the investigated conditions involved changes in fundamental cellular processes associated with protein synthesis. Such changes may reflect physiological adaptations that help *S. aureus* maintain cellular homeostasis under the investigated conditions.

Protein functional annotation using InterProScan confirmed that the representative AMR proteins contained conserved domains characteristic of their known biological functions. The identification of established protein families and functional domains supports the functional annotation of the representative AMR proteins and provides additional evidence that these proteins retain the structural features required for their respective resistance mechanisms. Protein interaction analysis further identified a functional association between GyrA and ParC, consistent with their complementary roles as subunits of bacterial type II topoisomerases involved in DNA replication (Mlynarczyk-Bonikowska et al., 2022). The remaining representative proteins did not form direct interactions within the selected reference proteome, indicating that no additional high-confidence functional associations were identified in the available STRING network. Together, the transcriptomic and protein-level analyses complemented the genomic findings by providing functional context for the representative AMR genes.

Comparison with Previous Studies

Previous genomic studies have described antimicrobial resistance in *Staphylococcus aureus* as a combination of conserved chromosomal determinants, acquired resistance genes, and resistance-associated mutations (John et al., 2019; Pennone et al., 2022). The comparative genomic findings of this study support this model. Several efflux pump genes and regulatory determinants were classified as components of the core genome, suggesting that they represent stable features of the analysed *S. aureus* genomes. In contrast, *mecA* and its associated regulatory genes showed variable distributions and were assigned mainly to the shell genome. This agrees with previous reports that a substantial proportion of acquired resistance genes occur within the accessory genome, while more widely conserved genes may contribute to intrinsic or basal resistance-related functions (John et al., 2019; Pennone et al., 2022). Previous studies have described *S. aureus* as having an open pan-genome, which may

facilitate continued variation in its resistance gene repertoire as additional genes are acquired or lost between lineages.

Methicillin resistance has been widely associated with the acquisition of *mecA*, which encodes the altered penicillin-binding protein PBP2a and reduces susceptibility to β -lactam antibiotics (Guo et al., 2020; Mlynarczyk-Bonikowska et al., 2022; Naorem et al., 2020). The detection of *mecA*-carrying SCCmec elements in the three assembled SRA genomes is consistent with this established mechanism. The occurrence of different SCCmec types among these genomes also indicates variation in the genetic background carrying methicillin resistance. Previous studies have identified SCCmec, plasmids, transposons and insertion sequences as important vehicles for the mobility of resistance determinants (John et al., 2019; Pennone et al., 2022; Souza et al., 2024). In the current study, MobileElementFinder detected several insertion sequences and transposable elements, supporting their potential contribution to genome plasticity and resistance-gene dissemination. However, the AMR genes examined were not located within the genomic islands predicted by IslandViewer. This suggests that SCCmec and other detected mobile elements were more relevant to the observed resistance context than the predicted genomic islands, although direct horizontal transfer events were not demonstrated.

The transcriptomic findings are also relevant to a limitation commonly reported in genomic AMR studies: the presence of a resistance gene does not necessarily establish whether it is expressed or whether it produces a resistant phenotype. Gene regulation, expression levels and other physiological responses can influence the relationship between genotype and antimicrobial susceptibility (Babiker et al., 2019; Mason et al., 2018). The RNA-seq analyses in this study partly addressed this limitation by examining transcriptional changes under three experimental settings. GO enrichment of the PRJNA1358579 dataset indicated that the upregulated annotated genes were mainly associated with ribosomal structure and translation. These findings point to changes in fundamental cellular activity rather than enrichment of a specific resistance pathway. They therefore complement, but do not independently confirm, the resistance potential predicted from genomic data. Phenotypic susceptibility testing would still be required to establish whether the detected genes and transcriptional changes produced measurable antimicrobial resistance.

Previous literature has also emphasized the value of protein annotation and network-based approaches for examining the functions and relationships of resistance determinants (Hasan et al., 2025; Jones et al., 2014; Szklarczyk et al., 2023). In this study, InterProScan identified protein families and conserved domains consistent with the expected functions of the selected AMR proteins, providing functional support for the sequence-based gene predictions. STRING analysis produced a more limited interaction pattern than might be expected from a broadly interconnected resistance network. A functional association was identified between GyrA and ParC, which is consistent with their related roles in bacterial DNA topology, whereas the other mapped proteins remained isolated in the selected reference proteome. This indicates that only limited high-confidence interactions were identified among the analysed proteins in the selected reference proteome. These findings are consistent with the multifactorial nature of AMR, in which resistance determinants can contribute through distinct biological mechanisms rather than a single interaction network. Thus, the findings support the literature describing AMR as a multifactorial trait, while also showing that direct protein interactions among selected resistance determinants may be limited.

Strengths and Limitations of the Study

A major strength of this study is the integration of multiple bioinformatics approaches to investigate antimicrobial resistance in *Staphylococcus aureus*. Rather than relying on a single analytical method, the workflow combined comparative genomics, transcriptome analysis, protein functional annotation, mobile genetic element analysis, and protein interaction analysis to examine antimicrobial resistance from complementary perspectives. The use of publicly available whole-genome and RNA-seq datasets enabled the investigation of both the genetic basis and the functional characteristics of antimicrobial resistance. Furthermore, integrating the outputs from RGI, Panaroo, SCCmecFinder, IslandViewer, MobileElementFinder, InterProScan, and STRING into a single evidence table provided a comprehensive framework for interpreting the resistance determinants identified during the study.

Despite these strengths, several limitations should be acknowledged. The analyses were entirely computational and relied on publicly available sequencing datasets and curated reference databases. Consequently, the findings represent predictions based on existing genomic and functional annotations and were not validated by experimental approaches such as antimicrobial susceptibility testing, quantitative gene expression assays, or protein functional studies. In addition, the identification of resistance genes depends on currently available database entries, meaning that novel or previously uncharacterized resistance mechanisms may not have been detected (**Babiker et al., 2019; Mason et al., 2018; Pennone et al., 2022**).

Another limitation is that the genomic and transcriptomic datasets were obtained from independent studies conducted under different experimental conditions. Although each dataset contributed valuable information, direct comparisons between genomic variation and gene expression patterns should be interpreted with caution because they were not generated from the same bacterial isolates. Similarly, the protein interaction analysis was constrained by the available reference proteome, limiting the number of high-confidence interactions that could be identified among the representative AMR proteins. Therefore, the interaction network presented in this study should be considered a useful functional reference rather than a complete representation of resistance-associated protein interactions.

Overall, the integrated bioinformatics workflow provided a systematic and comprehensive assessment of antimicrobial resistance in *Staphylococcus aureus*. While experimental validation would be required to confirm the biological significance of several computational predictions, the study demonstrates that combining multiple complementary analyses provides a broader understanding of the genetic and functional basis of antimicrobial resistance.

CONCLUSION

This study employed an integrated bioinformatics workflow to investigate antimicrobial resistance (AMR) in *Staphylococcus aureus* by combining comparative genomics, transcriptome analysis, protein functional annotation, and genomic context analysis. Using publicly available whole-genome and RNA sequencing datasets, the study examined antimicrobial resistance from multiple complementary perspectives to better understand the distribution and functional characteristics of resistance determinants. The findings demonstrate that integrating genomic and transcriptomic analyses provides a more comprehensive understanding of AMR than relying on any single analytical approach alone.

A key outcome of this study was the development of an integrated bioinformatics workflow for investigating antimicrobial resistance in *S. aureus*. Comparative genomic analysis distinguished conserved resistance determinants from more variably distributed genes, providing insights into the evolutionary organization of the *S. aureus* resistome. The incorporation of transcriptomic analysis added a functional dimension by examining changes in gene expression under different experimental conditions, while protein functional annotation and interaction analyses provided additional evidence supporting the predicted roles of representative AMR proteins. Integrating these complementary analyses into a single evidence framework facilitated a comprehensive interpretation of antimicrobial resistance.

The integrated workflow developed in this study provides a systematic framework for investigating antimicrobial resistance using complementary bioinformatics approaches. By combining outputs from resistance gene identification, pan-genome analysis, transcriptomics, protein functional annotation, and mobile genetic element analyses, it was possible to evaluate antimicrobial resistance from both genetic and functional perspectives. Although the findings are based entirely on computational analyses and require experimental validation, the workflow provides a systematic framework that can be applied to future genomic investigations of bacterial antimicrobial resistance.

Future studies could extend this work by incorporating larger collections of clinical isolates, long-read genome sequencing, phenotypic antimicrobial susceptibility testing, and additional multi-omics datasets to strengthen genotype-phenotype relationships and improve the characterization of resistance mechanisms. Overall, this study demonstrates that an integrated bioinformatics approach provides a robust and comprehensive strategy for investigating antimicrobial resistance in *Staphylococcus aureus* and highlights the role of computational biology in advancing antimicrobial resistance research and genomic surveillance.

SUPPLEMENTARY DATA

Supplementary Tables S1–S9 are provided as a separate Excel workbook accompanying this report. The workbook contains dataset metadata, genome assembly statistics, CARD-RGI findings, differential-expression results, pangenome analysis, genomic-context analyses, protein functional annotations, GO enrichment results, and the integrated AMR evidence table.

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